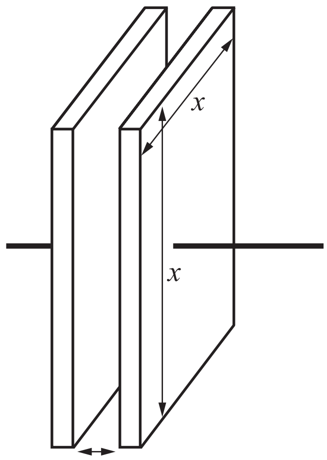
Examiner only

SECTION A

Answer **all** *questions.*

1. *(a)* (i) An 8.0 nF capacitor has square plates separated by 0.20 mm of air. Calculate the length (x) of the sides of the plates. [3]



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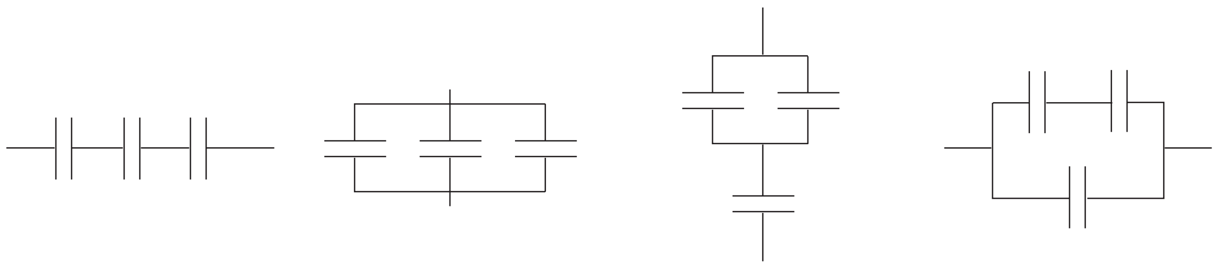
(ii) Which of the following combinations of 8.0 nF capacitors has a capacitance of 12.0 nF? Justify your answer. [4]

All individual capacitors shown are 8.0 nF.



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(iii) State how the capacitance of the capacitor can be increased without changing its dimensions. [1]

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(iv) Calculate the charge on the plates of the 8.0 nF capacitor when it stores an energy of 0.62 μJ. [3]

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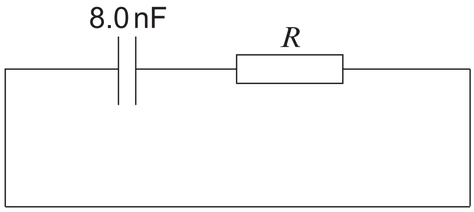
(b) (i) A charged 8.0 nF capacitor is discharged through a large resistor. Calculate the resistance of the resistor if the time constant is 15.5 ms. [2]



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(ii) Calculate the fraction of the initial charge remaining on the capacitor after 31.0 ms. [3]

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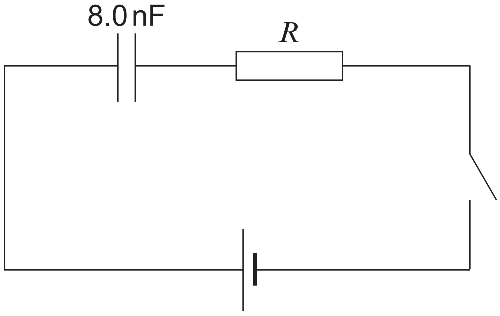
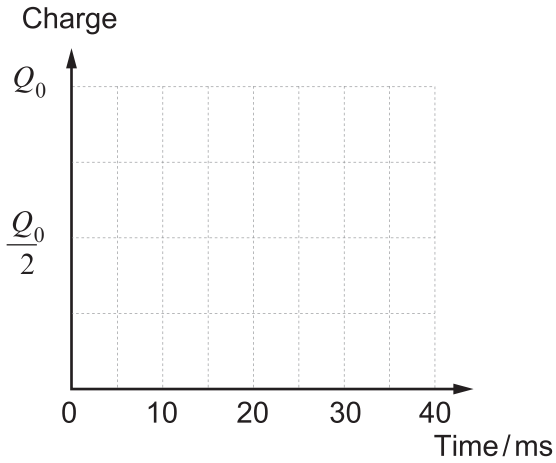
(c) An identical, uncharged 8.0 nF capacitor is **charged** in the following circuit, **using the same large resistor, R**. The switch is closed and the capacitor is uncharged at time $t = 0$.

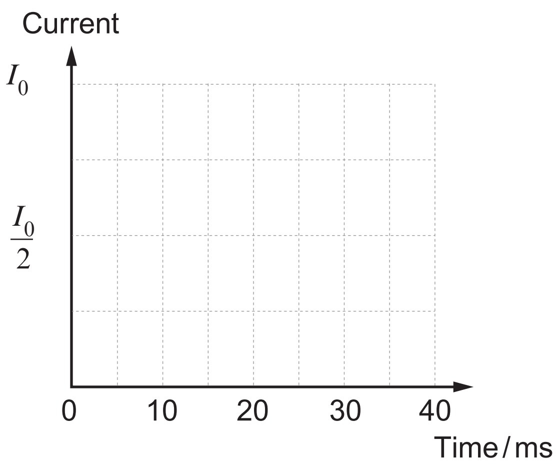


Without further calculations, sketch graphs of:

(i) the charge on the plates of the capacitor against time; [2]



(ii) the current in the circuit against time. [2]



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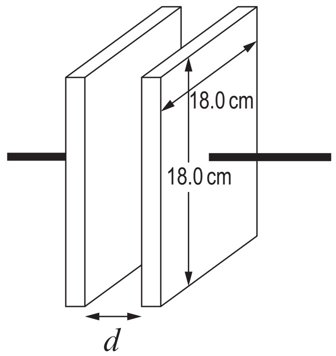
SECTION A

Answer **all** *questions.*

1. *(a)* (i) A laboratory technician has two square aluminium plates with sides of length 18.0 cm. Calculate the separation, d, of the plates that produces a capacitance of 2.0 nF. [3]



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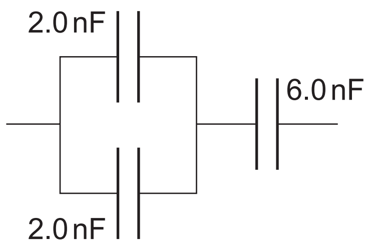
(ii) Calculate the capacitance of the capacitor combination shown. [3]



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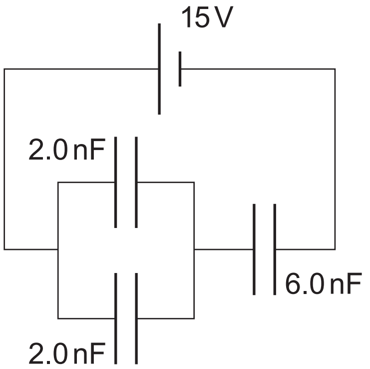
(iii) Show that the charge stored on a 2.0 nF capacitor in the following circuit is 18 nC. [3]



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(iv) Calculate the total energy stored by the capacitor combination. [2]

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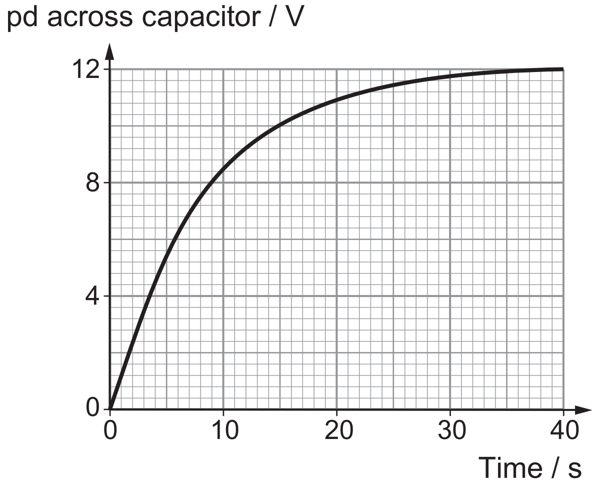
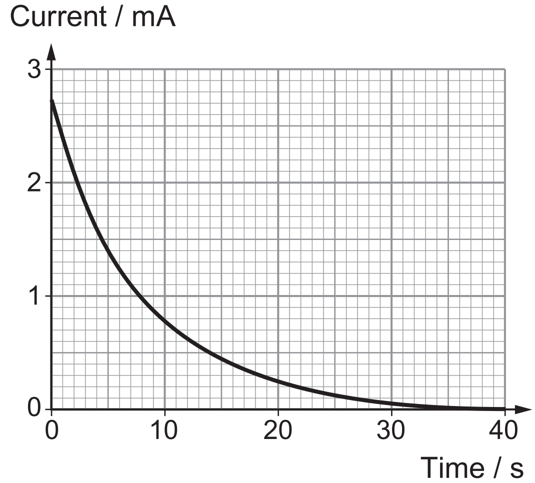
(b) A group of students investigated the **charging** of a capacitor through a resistor and obtained the following data.





(i) Draw a circuit diagram of the apparatus that might have been used to obtain the data. [2]

(ii) Use the graphs to determine:

- the emf of the power supply;
- the resistance of the resistor;
- the capacitance of the capacitor. [5]

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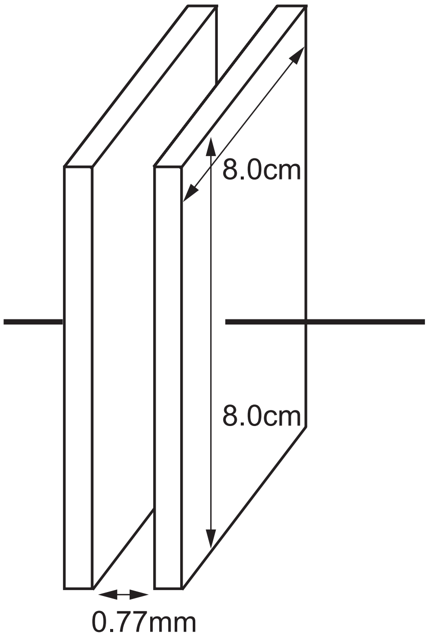
SECTION A

*Answer **all** questions.*

1. *(a)* (i) For the air-spaced parallel plate capacitor shown, calculate the pd applied when it stores 25.5 μJ of energy. [4]



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(ii) Explain why the capacitor stores energy when a pd is applied to the plates. [2]

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(iii) A group of scientists claim that they have developed a new dielectric that enables the above capacitor to store a million times more charge and energy for a given pd. Explain what further steps must be taken by the scientific community and industry before this new dielectric can be used in devices and sold to the public. [3]

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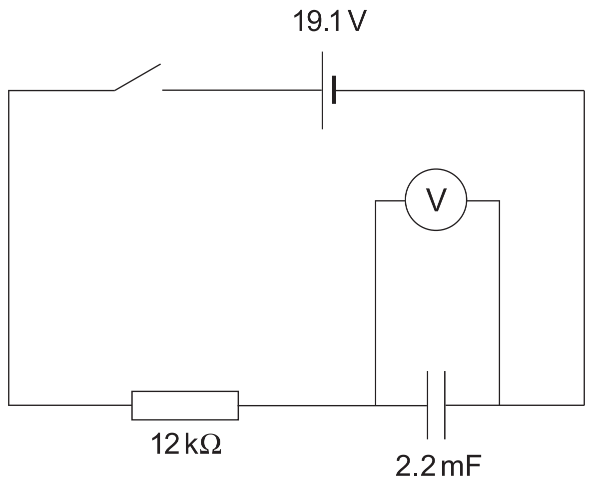
(b) Bethan investigates the charging of a capacitor using the following circuit.



The results she obtains are then tabulated.

Time / s	pd across capacitor / V ± 0.5 V	Charge on capacitor / mC ± 1.1 mC
10.0	6.1	13.4
20.0	10.0	22.0
30.0	12.8	....
40.0	15.1	33.2
50.0	16.2	35.6
60.0	17.1	....
70.0	17.7	38.9
80.0	18.3	40.3

(i) Confirm that the uncertainty in the charge is 1.1 mC (you may assume that the uncertainty in the 2.2 mF capacitor is negligible). [1]

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(ii) **Complete** the table. [2]

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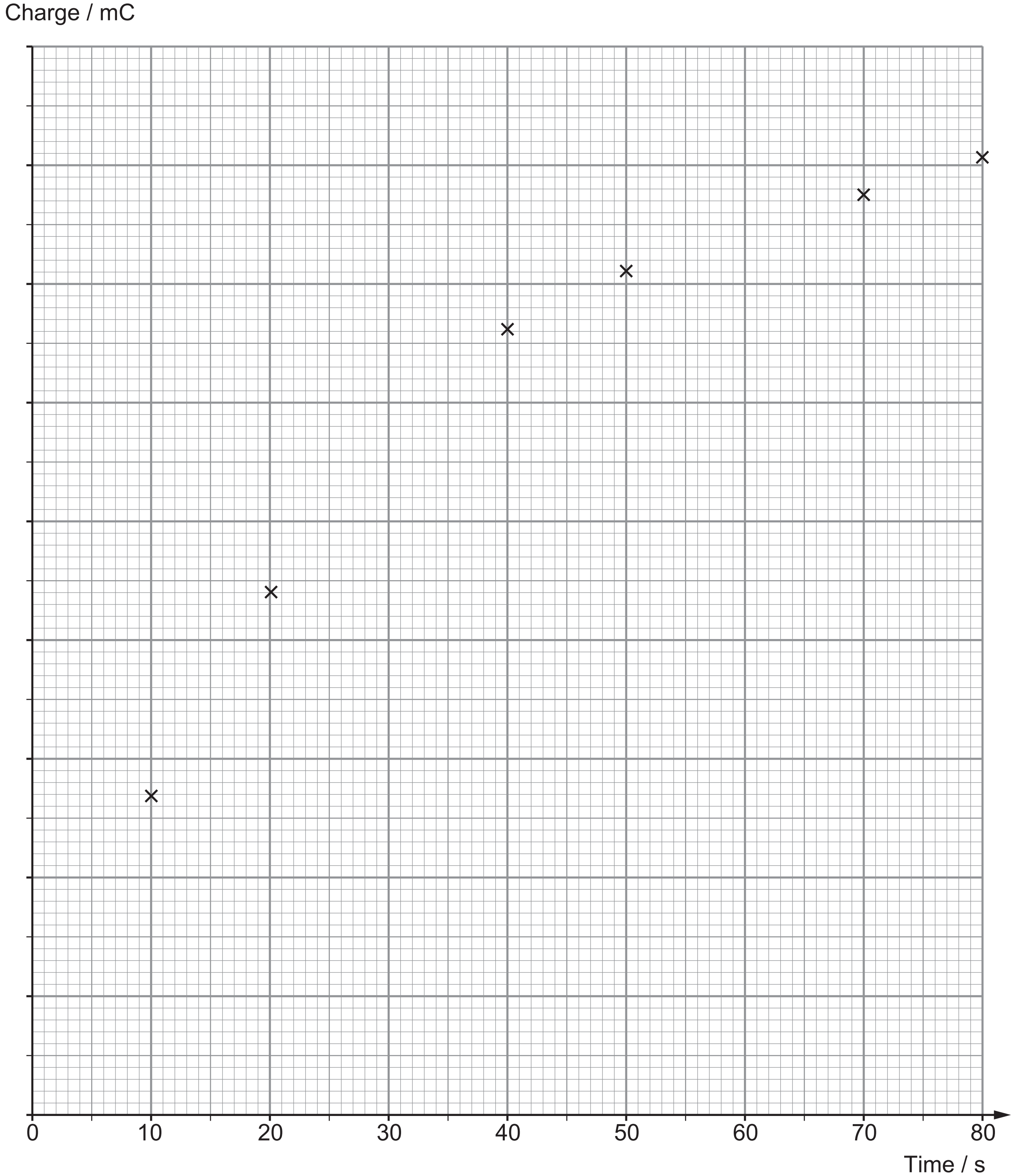
(iii) Complete the graph by labelling the y-axis scale, plotting the remaining two points, adding error bars for charge and drawing a curve of best fit. [5]



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(iv) **Use the curve of best fit** to obtain the time constant of the charging circuit (show your working and do not simply multiply the resistance by the capacitance). [3]

(v) By drawing a suitable tangent, calculate the current in the circuit at 45 s. [2]

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(c) Use the graph on page 5 and your answers to *(b)*(iv) and *(b)*(v) to evaluate whether or not the data obtained in this experiment are in good agreement with the equations: [5]

$$Q = Q_0\left(1 - e^{-\frac{t}{RC}}\right) \textbf{ and } I = I_0 e^{-\frac{t}{RC}}$$

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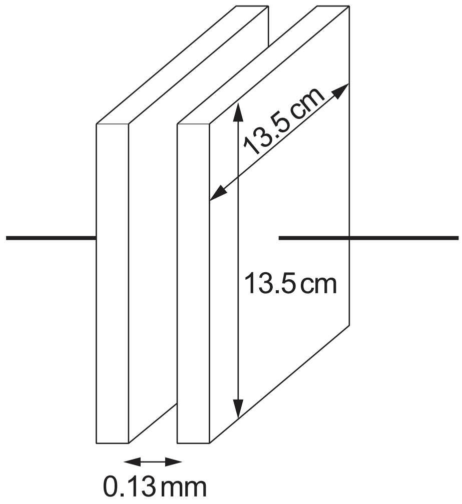
4. (a) Calculate the capacitance of the capacitor shown. [2]



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(b) (i) Calculate the pd across the terminals of a 5.0 mF capacitor when it stores 1.00 J of energy. [2]

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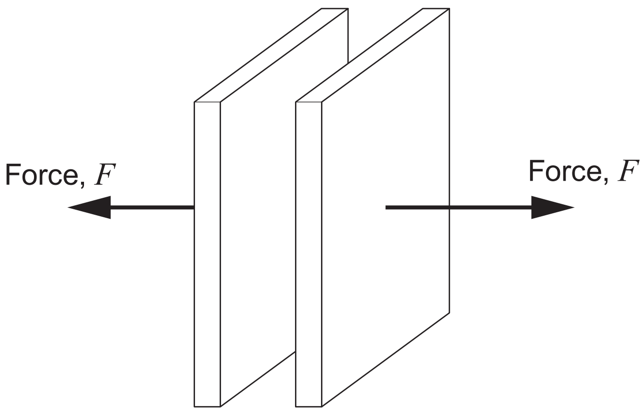
(ii) In the space provided, draw a diagram to show how you would combine three 5.0 mF capacitors to produce a capacitance of 7.5 mF. [2]

(c) The separation of the plates of a charged capacitor is increased by the application of a force, F. The capacitor is **isolated** so the charges on the plates remain unchanged.



(i) State why a force must be exerted to separate the charged plates. [1]

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(ii) Explain why the capacitor stores more **energy** when the separation of the plates is increased even though the charge remains constant. [2]

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(iii) Show that the energy stored by the capacitor is given by:

$$\text{energy stored} = \frac{1}{2}\frac{Q^2 d}{\varepsilon_0 A}$$

where Q is the charge stored, d is the separation of the plates, ε_0 is the permittivity of free space and A is the area of the plates. [2]

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(iv) Bethan states that the force, F, required to separate the plates is given by:

$$F = \frac{1}{2}\frac{Q^2}{\varepsilon_0 A}$$

Determine whether Bethan is correct to arrive at this conclusion. [2]

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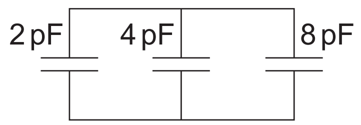
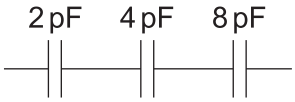
SECTION A

Answer **all** questions.

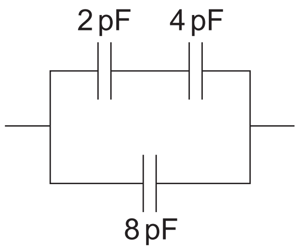
1. (a) Calculate the total capacitance of each of the following capacitor combinations. [6]



(i)



(ii)



(iii)

(i) ....

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(ii) ....

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(iii) ....

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(b) Michael, a laboratory technician, must make a 2.7 µF capacitor using only two square sheets of aluminium of side 42.0 cm. Determine whether or not it is realistic for Michael to produce a capacitance of 2.7 µF using these two sheets of aluminium and no dielectric. [4]

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(c) State what happens to the capacitance of a parallel plate capacitor when a dielectric is placed between the plates. [1]

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Examiner only

SECTION A

Answer **all** questions.

1. A laboratory technician needs to make a 230 pF capacitor using two square metal plates of side length 8.7 cm.

(a) Calculate the separation of the metal plates in order to produce this capacitance. [3]

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(b) The technician states that the 230 pF capacitor stores approximately 0.9 μJ of energy when it stores a charge of 20 nC on its plates. Determine whether or not the technician is correct. [3]

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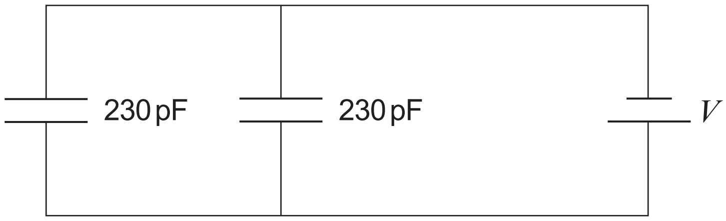
(c) By discussing the pd and charge on each of the capacitors, explain why the capacitance of the two 230 pF capacitors connected in parallel is 460 pF. [3]



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(d) **Draw a diagram** to show how you could combine three capacitors of 230 pF to provide a capacitance of approximately 150 pF. [1]

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2. Describe how you would carry out an experiment to investigate the discharging of a capacitor through a resistor to determine the time constant. [6 QER]

6